CLOUD AND SERVERLESS COMPUTING

PROJECT ABSTRACT

TITLE: CLOUD-BASED ATTENDANCE SYSTEM USING QR

Traditional attendance systems, including manual registers and biometric devices, often suffer from inefficiencies such as time consumption, data inaccuracies, proxy attendance, and dependency on physical infrastructure. These limitations become more significant in large institutions where managing attendance for a high number of users is challenging. Hence, there is a need for a reliable, scalable, and contactless solution that ensures accuracy and real-time accessibility of attendance data.

The objective of this project is to design and develop a **cloud-based attendance system using QR codes** that enables seamless and secure attendance tracking. The system aims to eliminate manual intervention, reduce fraudulent practices, and provide a scalable solution capable of handling large numbers of users efficiently.

The proposed system is built using a **serverless cloud architecture** on platforms such as Amazon Web Services or Microsoft Azure. It utilizes services like AWS Lambda for backend processing, Amazon API Gateway for API management, and Amazon DynamoDB for scalable data storage. Authentication and access control are handled using Amazon Cognito, ensuring secure user management. QR codes are dynamically generated for each session and scanned by users through a web or mobile interface.

The system allows administrators to generate QR codes for attendance sessions, while users can scan these codes to mark their presence in real time. The backend validates the QR data and records attendance securely in the cloud database.

The proposed solution enhances efficiency, ensures data accuracy, prevents proxy attendance, and provides a scalable and cost-effective attendance management system suitable for educational institutions and organizations.

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